Saba Art

Handmade resin art · Thrissur, Kerala



FUNCTIONAL REQUIREMENTS

AI Messaging Assistant

Functional Requirements Document

PREPARED FOR	PREPARED BY	VERSION	DATE	STATUS
Saba Art Thrissur, Kerala	**Tecza** Development team	**1.0**	**2 October 2026**	**Draft for review**

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1 Document control

Version history

Version	Date	Author	Changes
1.0	2 October 2026	Tecza	Initial draft for review.

Reviewers and approvers

Role	Name	Responsibility
Owner, Saba Art	[To be completed]	Reviews and approves the business requirements, rules and acceptance scenarios.
Project lead, Tecza	[To be completed]	Reviews and approves technical feasibility, scope and phasing.

Distribution

- Saba Art — owner.
- Tecza — project team (development and testing).

2 Introduction

2.1 Purpose

This document defines what the Saba Art AI Messaging Assistant must do. It is the agreed reference for the client and the development team when designing, building and accepting the system. Each requirement has a unique ID, a priority and a delivery phase so that it can be traced through development and testing.

Priorities: Must required for the phase to be accepted; Should important, delivered in the phase unless agreed otherwise; Could desirable, delivered if time allows.

2.2 Scope

The scope covers two connected parts:

- **The AI messaging assistant** — receives customer messages from WhatsApp and Instagram, answers routine questions automatically from the business's own information, and hands everything sensitive to a person.
- **The admin and staff dashboard** — a web app, installable on a phone, where the owner and helpers read conversations, reply, handle escalations and manage the business information the AI uses.

Work is planned in four phases, as set out in the project proposal. The phase shown against each requirement indicates when it is delivered.

Phase	Focus	Summary
1	Working demo and setup	AI replies to routine questions, hand-over to the owner or helpers, the admin and staff dashboard, knowledge base editing, analytics, audit log and demo mode with simulated customers.
2	Go live on WhatsApp and Instagram	Connection of the real WhatsApp Business number and Instagram account through Meta (including business verification and app approval), replies to real customers, and alerts by phone notification and email.
3	Orders and customer tools	Order board, payment records, message templates, broadcasts, customer lists and product catalogue.
4	Payments and delivery tracking	Payment links, shipping and tracking, Instagram comment replies, away message and 24-hour window alerts.

2.3 Glossary

Term	Meaning
Escalation	Handing a conversation from the AI to a person (admin or staff) because it needs human judgement. Every escalation has a recorded reason.
Auto-reply	A reply written by the AI and sent to the customer without human review. Allowed only for approved routine questions.
Knowledge Base (KB)	The business's own information that the AI may use: products, prices, custom-order rules, delivery, payment, policies, tone examples and always-escalate topics.
24-hour window	The 24 hours after a customer's most recent message, during which the business may send free-form replies. Outside it, WhatsApp only allows approved template messages.
Echo	A reply the owner sends directly from the WhatsApp or Instagram phone app, which Meta reports back to the system through the webhook.
Demo mode	An operating mode in which incoming messages are simulated and outgoing replies are logged instead of being sent through Meta.
Batching	Waiting 10 seconds so that several messages a customer sends in quick succession are answered with one reply.
Intent	The AI's classification of what the customer wants: `price`, `delivery`, `care`, `custom_order`, `payment`, `complaint`, `refund`, `unknown` or `off_topic`.
Conversation state	Who is currently responsible for a conversation: the AI (`ai_active`), waiting for a person (`escalated`), a person (`owner_handling`), or automation stopped (`paused`).
Admin	The business owner. One per business, with full access to the dashboard.
Staff	A helper employed by the business who handles escalated conversations only.
PWA	Progressive Web App: a website that can be added to a phone's home screen and opened full-screen like an app, without an app store.
Template message	A WhatsApp message format approved in advance by Meta. Required for business-initiated messages outside the 24-hour window; Meta charges for these messages.

3 Business context and objectives

3.1 Context

Saba Art is a small handmade resin-art business in Thrissur, Kerala. Customers contact the business on WhatsApp and Instagram. Most messages are repetitive questions — prices, delivery times and charges, product care and custom orders — written in English, Malayalam or Manglish (Malayalam written in English letters). Answering these by hand takes the owner's time away from making products, and customers who message outside working hours wait for a reply.

A smaller number of messages are sensitive: refunds, complaints, payment claims, damaged items and questions about products the business does not list. These must always be handled by a person.

3.2 Objectives

1. **Instant replies, 24/7, for routine questions**, grounded only in the business's own price list and policies.
2. **Never let the AI handle sensitive matters.** Refunds, complaints, payment claims and unknown products go to a person.
3. **One dashboard** for the owner and helpers, usable on a phone.
4. **A full audit trail** of every reply, escalation and change.

3.3 Success measures

Targets will be agreed with the owner before go-live (see Open Questions).

Measure	Definition	Target
Auto-reply rate	Share of customer message batches answered by the AI without a person.	To be agreed
Average response time	Average time from a customer's message to the first reply (AI or person).	To be agreed
Escalations resolved	Share of escalations resolved, and the time taken to resolve them.	To be agreed
Wrong prices sent	Number of AI replies that state a price or fact that is not in the knowledge base.	None

4 Users and roles

4.1 Actors

Actor	Description	Interaction with the system
Customer	A person who messages Saba Art on WhatsApp or Instagram.	Uses only WhatsApp or Instagram. Never sees the dashboard.
Admin	The business owner. One admin per business.	Full access to every part of the dashboard.
Staff	Helpers hired by the business. Several staff per business.	Handles escalated conversations only.
AI assistant	Google Gemini, guided by the knowledge base and the business rules.	Answers routine questions and creates escalations.
System	Automated background processes: job queue, reminders, token refresh and notifications.	Acts without a user; recorded as "system" in the audit log.

4.2 Permission matrix

Capability	Admin	Staff
View all conversations (every state and channel)	✓ Yes	✗ No
View escalated conversations assigned to them or unassigned	✓ Yes	✓ Yes
Reply to customers	✓ Any conversation	○ Only those escalations
Resolve a conversation (return it to the AI)	✓ Yes	○ Only those escalations
Assign escalations to staff	✓ Yes	✗ No
Manage staff accounts	✓ Yes	✗ No
Edit the knowledge base	✓ Yes	✗ No
Pause and resume automation	✓ Yes	✗ No
View analytics	✓ Yes	✗ No
View the audit log	✓ Yes	✗ No
Settings and channel connections (WhatsApp, Instagram)	✓ Yes	✗ No

Access principle

Staff see the minimum needed to handle the customer in front of them, and nothing more. Roles are enforced by the backend on every request, using the role in the verified login token. The dashboard hides pages a user cannot use, but this is for convenience only and is never relied on for security.

5 System overview

The diagram shows how a customer message travels through the system and how people interact with it.

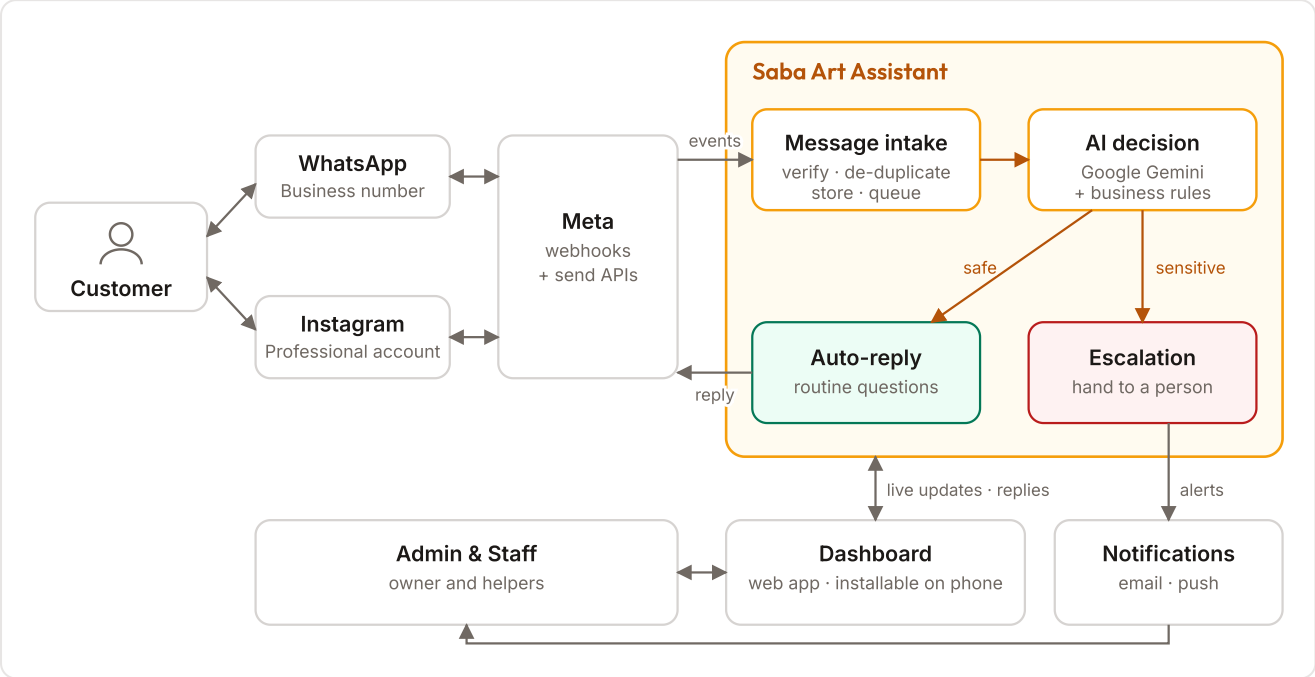


Figure 1 — System context

Channels	Customers use their normal WhatsApp or Instagram app. The business connects a WhatsApp Business number (WhatsApp Cloud API) and an Instagram Professional account (Instagram API with Instagram Login).
Meta webhooks	Meta delivers every new message and event to the assistant, and the assistant sends replies back through Meta's APIs.
Message intake	Checks that each event really comes from Meta, ignores duplicates, detects echoes, stores the message and a copy of any media, and places it in a job queue for processing.
AI decision	After a 10-second batching wait, sends the knowledge base, recent history and the new messages to Google Gemini (model `gemini-2.5-flash`) through the Gemini API. The result is checked against a strict format, then the business rules are applied.
Auto-reply or escalation	Routine questions are answered automatically. Anything sensitive, unclear or not covered by the knowledge base becomes an escalation for a person.
Dashboard	A web app for the admin and staff, installable on a phone as a PWA, with live updates of new messages and escalations.
Notifications	Push notifications and email alerts tell the admin about new escalations.
Data	A PostgreSQL database holds all records and the job queue. Redis is used for login rate limiting and for caching the knowledge base.

6 Functional requirements

Each requirement is written as "The system shall...". Priority and phase are defined in Section 2.

6.1 Message intake (INT)

ID	Requirement	Priority	Phase
FR-INT-01	The system shall receive incoming customer messages from WhatsApp and Instagram through Meta webhooks.	Must	2
FR-INT-02	The system shall complete Meta's webhook verification handshake, returning the challenge only when the verify token matches the configured value.	Must	2
FR-INT-03	The system shall validate the `X-Hub-Signature-256` header of every webhook event against the raw request body and shall reject any event whose signature is missing or invalid.	Must	2
FR-INT-04	The system shall acknowledge each webhook event immediately with HTTP 200 and process it asynchronously in a job queue.	Must	2
FR-INT-05	The system shall ignore any message whose Meta message ID has already been stored, so that redelivered events are never processed twice.	Must	2
FR-INT-06	The system shall detect echoes (replies the owner sends from the phone app), store them as owner messages, and switch the conversation to human handling (`owner_handling`) before any AI job runs for that conversation.	Must	2
FR-INT-07	The system shall download media attached to an incoming message immediately on arrival and keep its own copy, because Meta media URLs expire.	Must	2
FR-INT-08	The system shall keep one customer record per channel. The same person on WhatsApp and on Instagram is treated as two customers.	Must	1
FR-INT-09	The system shall keep one ongoing conversation per customer, holding the full message history.	Must	1

6.2 AI replies (AI)

ID	Requirement	Priority	Phase
FR-AI-01	The system shall process a message with the AI only when its conversation is in the `ai_active` state. In any other state the message is stored and not answered.	Must	1
FR-AI-02	The system shall wait 10 seconds after a customer message, combine all messages the customer sends in that time, and send at most one reply for the batch.	Must	1
FR-AI-03	The system shall ground every reply only in the knowledge base. The AI shall not state any price, product, time, charge or policy that is not in the knowledge base.	Must	1

ID	Requirement	Priority	Phase
FR-AI-04	For each batch, the AI shall return a decision containing: intent, reply text, facts used, missing facts, an escalate flag with reason, a suggested tag and the detected language.	Must	1
FR-AI-05	The system shall detect whether the customer writes in English, Malayalam or Manglish and reply in the same language.	Must	1
FR-AI-06	The system shall instruct the AI to follow the tone examples stored in the knowledge base.	Should	1
FR-AI-07	The system shall include up to the last 20 messages of the conversation as context for the AI.	Must	1
FR-AI-08	The system shall validate the AI output against a strict schema. If the output is invalid or the AI call fails, the system shall escalate with reason `parse_error` and shall never send a fallback or canned reply.	Must	1
FR-AI-09	The system shall send an auto-reply only when the intent is `price`, `delivery`, `care` or `custom_order` and no business rule requires escalation.	Must	1
FR-AI-10	The system shall record every AI decision, including whether its reply was sent.	Must	1
FR-AI-11	The system shall apply a 20-second timeout to each AI request and retry a limited number of times on temporary failures before escalating.	Must	1
FR-AI-12	The system shall use Google Gemini (`gemini-2.5-flash`) through the Gemini API. The model shall be configurable without code changes.	Must	1

6.3 Business rules (RUL)

ID	Requirement	Priority	Phase
FR-RUL-01	The system shall apply the business rules after every AI decision, in the order given in Section 7. The rules override the AI: they can turn an auto-reply into an escalation, never the reverse.	Must	1
FR-RUL-02	The system shall always escalate, with reason `always_escalate_phrase`, when the customer's message contains any of these phrases (case-insensitive): "i paid", "already paid", "payment done", "sent money", "where is my order", "not received", "damaged", "broken", "want refund", "cancel my order", "cheated", "fraud", "police", "consumer court".	Must	1
FR-RUL-03	The system shall always escalate the intents `complaint`, `refund`, `unknown`, `off_topic` and `payment`, with the reasons listed in Section 7.	Must	1
FR-RUL-04	The system shall escalate with reason `missing_facts` when the answer needs facts that are not in the knowledge base.	Must	1
FR-RUL-05	The system shall escalate with reason `unapproved_intent` any intent that is not in the auto-reply list (`price`, `delivery`, `care`, `custom_order`).	Must	1
FR-RUL-06	The AI shall never confirm that a payment has been received. Payment claims are always handled by a person.	Must	1

6.4 Escalations (ESC)

ID	Requirement	Priority	Phase
FR-ESC-01	The system shall create an escalation, set the conversation to `escalated` and record one reason: `intent_complaint`, `intent_refund`, `intent_unknown`, `missing_facts`, `always_escalate_phrase`, `payment_claim`, `unapproved_intent`, `parse_error`, `manual` or `window_closed`.	Must	1
FR-ESC-02	The system shall create new escalations unassigned, so that any staff member can pick them up.	Must	1
FR-ESC-03	The system shall allow the admin to assign an escalation to a staff member, recording who assigned it and when.	Must	1
FR-ESC-04	The system shall show the admin all open escalations, and show each staff member only the open escalations assigned to them or unassigned.	Must	1
FR-ESC-05	The system shall alert the admin when an escalation is created, by push notification and by email as a backup.	Must	2
FR-ESC-06	The system shall send a reminder when an escalation has had no reply from a person for 2 hours.	Should	2
FR-ESC-07	The system shall allow the admin, or a staff member permitted on that escalation, to resolve it. Resolving closes the escalation, records who resolved it and returns the conversation to `ai_active`.	Must	1
FR-ESC-08	The system shall record every escalation action (creation, assignment, reminder, resolution) in the audit log.	Must	1

6.5 Conversation management (CNV)

ID	Requirement	Priority	Phase
FR-CNV-01	The system shall keep every conversation in exactly one state — `ai_active`, `escalated`, `owner_handling` or `paused` — and change state only as defined in Section 8.	Must	1
FR-CNV-02	When the admin or a staff member replies from the dashboard, the system shall send the reply to the customer, set the conversation to `owner_handling` and stop AI replies until the conversation is resolved.	Must	1
FR-CNV-03	The system shall allow the admin to pause and resume automation for the whole business. While paused, incoming messages are stored but the AI does not reply.	Must	1
FR-CNV-04	The system shall support the conversation tags `new_lead`, `custom_order`, `payment_pending`, `order_confirmed` and `follow_up`. The AI may suggest a tag, which is applied with its auto-reply.	Should	1
FR-CNV-05	The system shall track the 24-hour customer-service window for each conversation and show the time left in the dashboard, highlighted in red when less than 2 hours remain.	Must	1

ID	Requirement	Priority	Phase
FR-CNV-06	If the 24-hour window has already closed when a message is processed, the system shall not auto-reply and shall escalate with reason `window_closed`.	Must	1
FR-CNV-07	Outside the 24-hour window, the system shall allow only approved WhatsApp template messages to be sent.	Must	2
FR-CNV-08	The system shall record every conversation state change in the audit log.	Must	1

6.6 Dashboard (DSH)

ID	Requirement	Priority	Phase
FR-DSH-01	The system shall let admin and staff log in with email and password, then open the inbox for the admin and the escalation queue for staff.	Must	1
FR-DSH-02	The system shall let users log out, and shall end a login session after a configured period (default 7 days), returning the user to the login page.	Should	1
FR-DSH-03	The admin inbox shall list all conversations with state, tag, channel and last message, filterable by state, tag and channel.	Must	1
FR-DSH-04	The chat view shall show the full message thread, with a distinct colour for each sender: customer, AI, owner, staff and system.	Must	1
FR-DSH-05	The chat view shall include a reply box with Send and Resolve actions.	Must	1
FR-DSH-06	The staff escalation queue shall list the staff member's assigned and unassigned open escalations, with reason and waiting time.	Must	1
FR-DSH-07	Staff shall open a single chat view for one escalated conversation, showing its history and the customer profile.	Must	1
FR-DSH-08	The dashboard shall update in real time when new messages, escalations, assignments or state changes occur, without reloading the page.	Should	1
FR-DSH-09	If the real-time connection is lost, the dashboard shall poll for updates every 5 seconds until it reconnects.	Should	1
FR-DSH-10	The dashboard shall be installable on a phone as a PWA and open full-screen from the home screen.	Should	1
FR-DSH-11	After login, the dashboard shall ask permission for notifications and register the device to receive push notifications.	Should	2

6.7 Knowledge base (KB)

All knowledge base editing is restricted to the admin.

ID	Requirement	Priority	Phase
FR-KB-01	The system shall let the admin add, edit and remove products, each with name, price, sizes, lead time and custom options.	Must	1

ID	Requirement	Priority	Phase
FR-KB-02	The system shall let the admin edit the custom-order rules (for example bulk discount, extra lead time, and how to treat items not in the catalogue).	Must	1
FR-KB-03	The system shall let the admin edit delivery zones, each with area, delivery days, charge and availability.	Must	1
FR-KB-04	The system shall let the admin edit payment methods, the advance percentage and the balance terms.	Must	1
FR-KB-05	The system shall let the admin edit the policies for returns, cancellation, care and warranty.	Must	1
FR-KB-06	The system shall let the admin edit tone examples (a customer message paired with an ideal reply).	Should	1
FR-KB-07	The system shall let the admin edit the list of always-escalate topics given to the AI.	Should	1
FR-KB-08	Knowledge base changes shall take effect for the AI immediately; the cached copy shall be cleared on every save.	Must	1
FR-KB-09	The system shall record which admin last edited the knowledge base and when.	Should	1

6.8 User management (USR)

ID	Requirement	Priority	Phase
FR-USR-01	The system shall let the admin create staff accounts with name, email and password. The role of a created account is always staff.	Must	1
FR-USR-02	The system shall send a new staff member a welcome email with the login address and their login details.	Should	2
FR-USR-03	The system shall list staff accounts with name, email, active status and last login time.	Must	1
FR-USR-04	The system shall let the admin deactivate and reactivate staff accounts. A deactivated user shall not be able to log in.	Must	1
FR-USR-05	The system shall store passwords only as one-way hashes and shall never show or return a password or password hash.	Must	1

6.9 Analytics (ANL)

ID	Requirement	Priority	Phase
FR-ANL-01	The system shall show the admin message counts (received and sent).	Should	1
FR-ANL-02	The system shall show the auto-reply rate.	Should	1
FR-ANL-03	The system shall show the escalation rate.	Should	1
FR-ANL-04	The system shall show the average response time.	Should	1
FR-ANL-05	The system shall let the admin filter all analytics by date range and by channel.	Should	1

6.10 Audit log (AUD)

ID	Requirement	Priority	Phase
FR-AUD-01	The system shall record every reply, auto-reply, escalation creation, assignment and resolution, conversation state change, user-management change and settings change.	Must	1
FR-AUD-02	Each audit entry shall include the actor (user, AI or system), the time, the action and its details, with links to the related conversation, escalation or user.	Must	1
FR-AUD-03	The system shall let the admin view the audit log, filter it by actor and action, and page through it.	Must	1

6.11 Settings and channels (SET)

ID	Requirement	Priority	Phase
FR-SET-01	The system shall provide the admin with a clearly visible pause / resume switch for automation.	Must	1
FR-SET-02	The system shall show the connection status of the WhatsApp and Instagram channels.	Should	2
FR-SET-03	The system shall store Meta access tokens encrypted, never in plain text.	Must	2
FR-SET-04	The system shall check the Instagram access token daily and refresh it before it expires.	Must	2

6.12 Demo mode (DEM)

ID	Requirement	Priority	Phase
FR-DEM-01	The system shall let an authorised user simulate an incoming message (customer name, channel, text and optional image) from the dashboard or the API.	Must	1
FR-DEM-02	In demo mode, the system shall log outgoing replies instead of sending them to Meta.	Must	1
FR-DEM-03	The system shall accept simulated messages only when demo mode is switched on.	Must	1

6.13 Future requirements (phases 3–4)

These requirements are planned after go-live. The phase allocation follows the roadmap in the project proposal. FR-STA-01 is not yet allocated to a phase (see OQ-08).

ID	Requirement	Priority	Phase
Orders			
FR-ORD-01	The system shall let the admin create an order from a conversation.	Should	3
FR-ORD-02	The system shall track each order through the statuses `pending`, `confirmed`, `in_production`, `ready`, `dispatched`, `delivered` and `cancelled`.	Should	3
FR-ORD-03	The system shall show an order board grouping orders by status.	Should	3
FR-ORD-04	The system shall notify the customer when their order status changes.	Should	3
Payments			
FR-PAY-01	The system shall record the 50% advance and the balance for each order.	Should	3
FR-PAY-02	The system shall let the admin mark a payment as paid manually.	Should	3
FR-PAY-03	The system shall create Razorpay payment links. Only the gateway's verified payment confirmation, or the admin, may mark a payment as paid.	Could	4
Delivery tracking			
FR-DLV-01	The system shall create Shiprocket shipments, store the tracking number and record shipment status updates.	Could	4
Customers and segments			
FR-CUS-01	The system shall group customers into segments: favourites, priority, recent buyers and new.	Should	3
FR-CUS-02	The system shall process messages from priority customers first during busy periods.	Could	3

ID	Requirement	Priority	Phase
Message templates			
FR-TPL-01	The system shall let the admin create message templates organised by category, with variables such as `{customerName}`, `{product}`, `{trackingId}`, `{amount}` and `{balanceAmount}` filled in when sent.	Should	3
Broadcasts			
FR-BRD-01	The system shall let the admin send approved WhatsApp templates to selected customer segments, schedule them, and view delivery statistics.	Should	3
FR-BRD-02	The system shall never send a broadcast to a customer who has opted out.	Should	3
Other			
FR-CAT-01	The system shall provide a product catalogue with photos, built from the knowledge base.	Should	3
FR-IGC-01	The system shall respond to Instagram comments matching keyword rules with a public reply or a direct message.	Could	4
FR-AWY-01	The system shall send an away message outside business hours.	Could	4
FR-WIN-01	The system shall alert the admin before a conversation's 24-hour window expires.	Could	4
FR-STA-01	The system shall provide detailed message statistics with CSV export.	Could	TBC

7 Business rules summary

Before the rules run, three checks apply: the conversation must be `ai_active` and automation not paused (otherwise the message is stored with no reply); the 24-hour window must be open (otherwise `window_closed`); and the AI output must be valid (otherwise `parse_error`). The rules are then applied to every AI decision in this order. The first rule that matches decides the outcome.

Step	Check	Escalation reason	Outcome
1	Message contains an always-escalate phrase (FR-RUL-02)	`always_escalate_phrase`	Escalate
2	Intent is an always-escalate intent (FR-RUL-03)	complaint → `intent_complaint` refund → `intent_refund` unknown → `intent_unknown` payment → `payment_claim` off_topic → `unapproved_intent`	Escalate
3	The answer needs facts not in the knowledge base (FR-RUL-04)	`missing_facts`	Escalate
4	Intent is not `price`, `delivery`, `care` or `custom_order` (FR-RUL-05)	`unapproved_intent`	Escalate
—	None of the above	—	Auto-reply

Note on rule order

Intent is checked before missing facts. A question about a product that is not in the catalogue, which the AI labels `unknown`, therefore escalates as `intent_unknown` rather than `missing_facts`. Either way, the customer receives no automatic answer and a person handles it.

8 Conversation state machine

Every conversation is in exactly one of four states. The AI only works on conversations in `ai_active`.

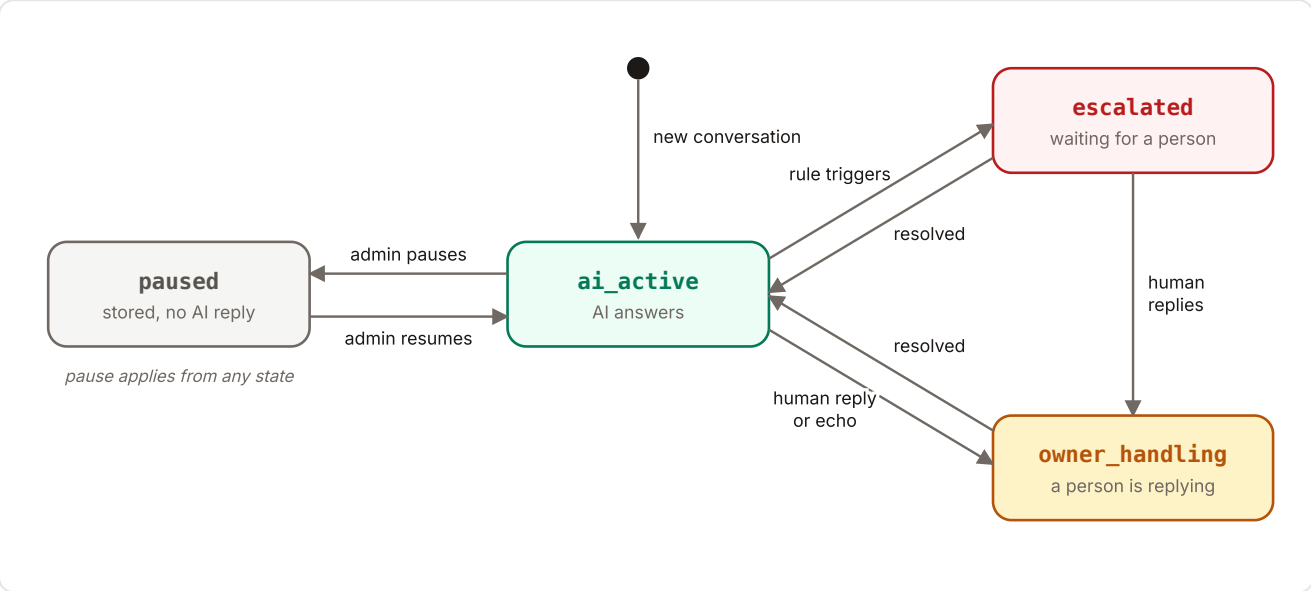


Figure 2 — Conversation states and transitions

From	Event	To	Who
ai_active	A business rule or AI decision requires a person	escalated	AI / system
ai_active	Admin or staff replies from the dashboard, or an echo arrives from the phone app	owner_handling	Admin / staff
escalated	A person replies to the customer	owner_handling	Admin / staff
escalated / owner_handling	Conversation marked resolved	ai_active	Admin / staff
Any state	Admin pauses automation	paused	Admin
paused	Admin resumes automation	ai_active	Admin

All state changes are written to the audit log. Staff cannot pause or resume automation. An echo must switch the conversation to `owner_handling` before any AI job runs for that conversation.

9 Acceptance scenarios

The following scenarios must pass, in demo mode, before Phase 1 is accepted.

9.1 Message handling

#	Customer message or action	Expected result	Covers
1	"How much is the ocean coaster set?"	Auto-reply stating ₹1,200.	FR-AI-09, FR-KB-01
2	"I want a custom clock with my daughter's name"	Auto-reply asking which colours the customer would like.	FR-AI-09
3	"Do you deliver to Bangalore?"	Auto-reply: 7–10 days, ₹120 shipping.	FR-AI-09, FR-KB-03
4	"How much for a 6 foot resin table?"	Escalated, reason `missing_facts`. No price is given.	FR-RUL-04
5	"I want a refund"	Escalated, reason `intent_refund`.	FR-RUL-03
6	"I paid already, where is my order?"	Escalated, reason `always_escalate_phrase`. The AI does not confirm the payment.	FR-RUL-02, FR-RUL-06
7	Three messages sent within 5 seconds	One combined reply, about 10 seconds after the last message.	FR-AI-02
8	Admin or staff replies from the dashboard	Conversation moves to `owner_handling`; the AI stops replying.	FR-CNV-02
9	"എത്ര രൂപ കൊടുക്കണം keychain ന്?"	Auto-reply in Malayalam stating ₹450.	FR-AI-05
10	Automation paused, then a message arrives	Message is stored; no AI reply is sent.	FR-CNV-03

9.2 Roles and security

#	Action	Expected result	Covers
R1	Staff member tries to open the Inbox, Settings, Users, Analytics, Audit log or Knowledge base pages	The dashboard redirects to the staff escalation queue.	Section 4.2
R2	Staff member calls the same admin functions directly through the API	Refused with HTTP 403 "Admin access required".	NFR-SEC-06
R3	Staff member opens an escalation assigned to another staff member	Refused with HTTP 403.	FR-ESC-04
R4	Six wrong login attempts from the same IP address within 15 minutes	The sixth attempt is blocked with HTTP 429.	NFR-SEC-05
R5	A deactivated staff member tries to log in	Login refused: "Account deactivated".	FR-USR-04

10 Non-functional requirements

ID	Category	Requirement
NFR-SEC-01	Security	Every dashboard and API request, except public endpoints (health check, login, webhook), shall require a valid signed login token (JWT).
NFR-SEC-02	Security	Passwords shall be hashed with argon2id.
NFR-SEC-03	Security	Stored Meta tokens shall be encrypted with AES-256-GCM.
NFR-SEC-04	Security	The signature of every webhook event shall be validated before it is processed.
NFR-SEC-05	Security	Login shall be limited to 5 attempts per 15 minutes per IP address; further attempts receive HTTP 429.
NFR-SEC-06	Security	Roles shall be enforced on the server for every request, using the role in the verified token, never a role sent by the browser.
NFR-SEC-07	Security	Production logs shall not contain message content; they record IDs only.
NFR-SEC-08	Security	Secrets and credentials shall never be committed to source control.
NFR-PRF-01	Performance	Webhook events shall be acknowledged immediately, before any processing.
NFR-PRF-02	Performance	A typical auto-reply shall be sent within about 10–20 seconds of the customer's last message, including the 10-second batching window.
NFR-AVL-01	Availability	The health endpoint shall be monitored every minute (BetterStack), with alerts to the development team if it fails.
NFR-AVL-02	Monitoring	Errors in the backend and dashboard shall be reported to an error-tracking service (Sentry).
NFR-SCL-01	Scalability	Every record shall carry a business ID, so the system can serve more than one business without redesign.
NFR-USE-01	Usability	The dashboard shall work on phone screens and be installable as a PWA.
NFR-USE-02	Usability	The dashboard interface shall be in English. The AI shall handle customers in English, Malayalam and Manglish.
NFR-PRV-01	Privacy	Customer data shall be used only to reply to and serve that customer.
NFR-PRV-02	Privacy	Customer media shall be stored as copies held by the system, not as links to Meta.
NFR-MNT-01	Maintainability	The system shall have an automated test suite covering the business rules, the AI pipeline and role-based access.

11 Integrations

Service	Used for	Phase
Meta WhatsApp Cloud API	Receiving and sending WhatsApp messages; template messages.	2
Instagram API with Instagram Login	Receiving and sending Instagram direct messages; comments later.	2
Google Gemini API	AI decisions and reply drafting (`gemini-2.5-flash`).	1
PostgreSQL database (Supabase)	All business records and the job queue.	1
Upstash Redis	Login rate limiting; knowledge base cache.	1
Resend	Escalation alert emails; staff welcome emails.	2
Web Push	Browser and phone notifications for escalations.	2
Sentry	Error tracking for backend and dashboard.	1
BetterStack	Uptime monitoring of the health endpoint.	1
Future		
Razorpay	Payment links with verified payment confirmation.	4
Shiprocket	Shipments, tracking numbers and status updates.	4
Cloudflare R2	Long-term storage of customer media.	To be scheduled

12 Assumptions, constraints and out of scope

12.1 Assumptions and constraints

- Meta business verification and App Review are required before going live. Their timing depends on Meta and is outside the project team's control.
- The WhatsApp number used for the business must not be registered on the WhatsApp phone app.
- The Instagram account must be a Professional account.
- Business-initiated WhatsApp messages outside the 24-hour window require approved template messages, and Meta charges for template messages.
- The quality of AI replies depends on an accurate, up-to-date knowledge base maintained by the business.

12.2 Out of scope (for now)

- Native apps in the Apple App Store or Google Play Store (the dashboard is installed as a PWA instead).
- International shipping.
- Taking payments inside the chat.
- Voice and video calls.

13 Open questions

#	Question	Owner	Status
OQ-01	The UPI ID and the demo login email addresses still use the old "asharesins" name. What are the correct values for Saba Art?	Saba Art	Open
OQ-02	What is the Instagram handle to be connected?	Saba Art	Open
OQ-03	Should staff be able to see and handle orders and payments once those features exist?	Saba Art	Open
OQ-04	Once orders exist, should the AI answer "where is my order?" from the order status? Today it always escalates.	Saba Art, Tecza	Open
OQ-05	What are the business hours, and what should the away message say?	Saba Art	Open
OQ-06	What targets should apply to the success measures in Section 3.3?	Saba Art, Tecza	Open
OQ-07	How long should messages, customer details and media be kept?	Saba Art, Tecza	Open
OQ-08	Detailed message statistics with CSV export (FR-STA-01) is not yet in the roadmap. Which phase should include it?	Saba Art, Tecza	Open

14 Approval

By signing below, the parties confirm that this document is the agreed basis for design, development and acceptance of the Saba Art AI Messaging Assistant.

Organisation	Name	Role	Signature	Date
Saba Art		Owner		
Tecza		Project lead		